
PORTA NUSA HOTEL

RFID Linen Visibility Platform

Technical Documentation & Operator Guide

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1. Executive Summary

Hotel linen management presents a persistent operational challenge: without item-level visibility, hotels cannot reliably track which linens have been dispatched to the laundry, which have returned, and which remain outstanding. Manual counting is error-prone, time-consuming, and provides no audit trail for reconciliation disputes.

The Porta Nusa Hotel RFID Linen Visibility Platform addresses this by attaching UHF RFID tags to individual linen items and reading them with a Chainway C5 E710 handheld scanner. Each scan session is uploaded in real time to a browser-based management interface, giving operators instant inventory counts, laundry batch tracking, and a complete transaction audit trail.

Dual-Mode Architecture

Simulation Mode	Hardware Mode
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The two modes are completely isolated at the database level. Simulation data never appears in Hardware Mode, and hardware scan data never appears in Simulation Mode. Reconciliation is computed automatically from live send and return counts — no manual tallying required. Every RFID session is permanently logged for audit purposes.

2. Solution Architecture

The platform integrates four layers: UHF RFID tags on linen items, the Chainway C5 E710 Android handheld, a Next.js 14 web application with REST API, and two isolated SQLite databases managed by Prisma ORM.

RFID Tags	Chainway C5 E710	Next.js 14 Web App	SQLite Databases
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Data flows left to right: tags → C5 → API → database.

Data Flow

Browser (Simulation Mode)	X-Demo-Mode: SIMULATION	Simulation database
Browser (Hardware Mode)	X-Demo-Mode: HARDWARE	Hardware database
Chainway C5 Android App	X-Demo-Mode: HARDWARE (always)	Hardware database

Key Technologies

Web Application	Next.js 14 (App Router)	UI and API route handlers
ORM	Prisma	Type-safe database access; two datasource instances
Databases	SQLite (two isolated files)	Simulation database and hardware database; no shared state

Android App	Kotlin / Android SDK	Scan sessions and RFID trigger handling
RFID Module	RFIDWithUHFUART SDK	UHF antenna control; EPC and RSSI capture
Language	TypeScript	Strict typing across the web codebase

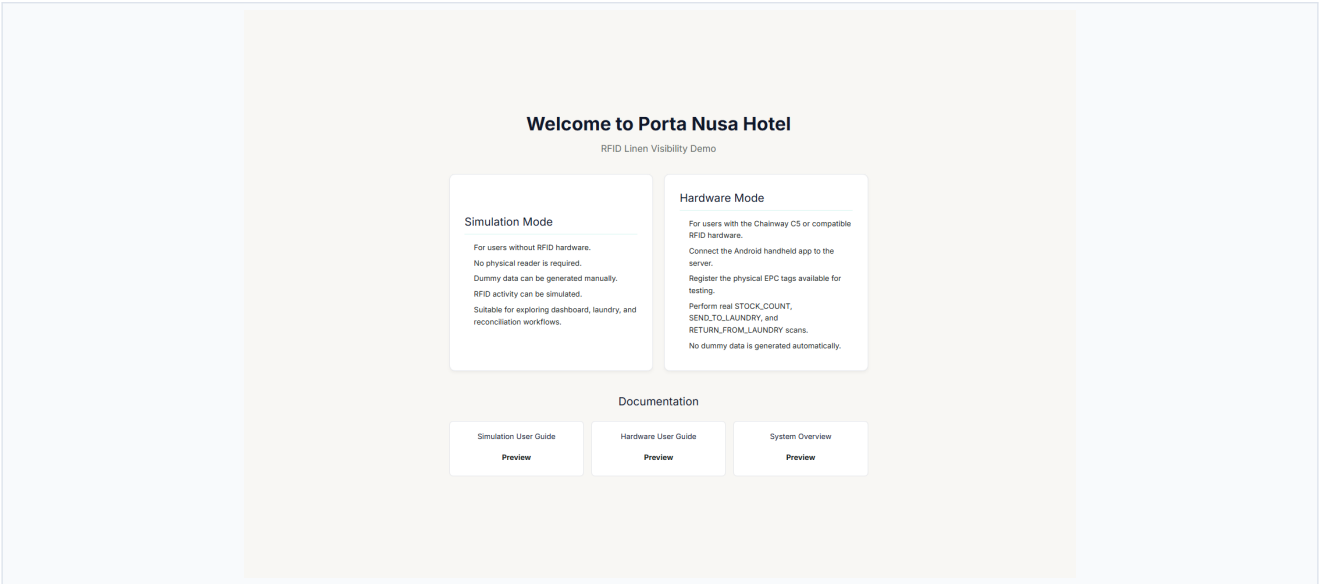
3. Operating Modes and Database Isolation

The browser operator selects the active mode using the sidebar toggle. All subsequent reads and writes target the selected mode's database exclusively. There is no data transfer between modes.

Populated by	Dashboard data-management actions	Chainway C5 Android uploads only
Linen item cap	100 (enforced)	100 (enforced)
Reset method	Dashboard Simulation Data Management	CLI scripts — no browser reset in MVP
Shared state	None	None

Mode Header: Browser mode is propagated via middleware that reads a cookie and attaches the X-Demo-Mode header to every server request. The Chainway C5 always sends X-Demo-Mode: HARDWARE regardless of browser state.

Mode Selection



Mode Selection Landing Page — the operator chooses Simulation or Hardware Mode before entering the platform.

4. End-to-End Hardware Workflow

The following sequence describes the complete physical linen lifecycle as executed using the Chainway C5 and the Hardware Mode web interface.

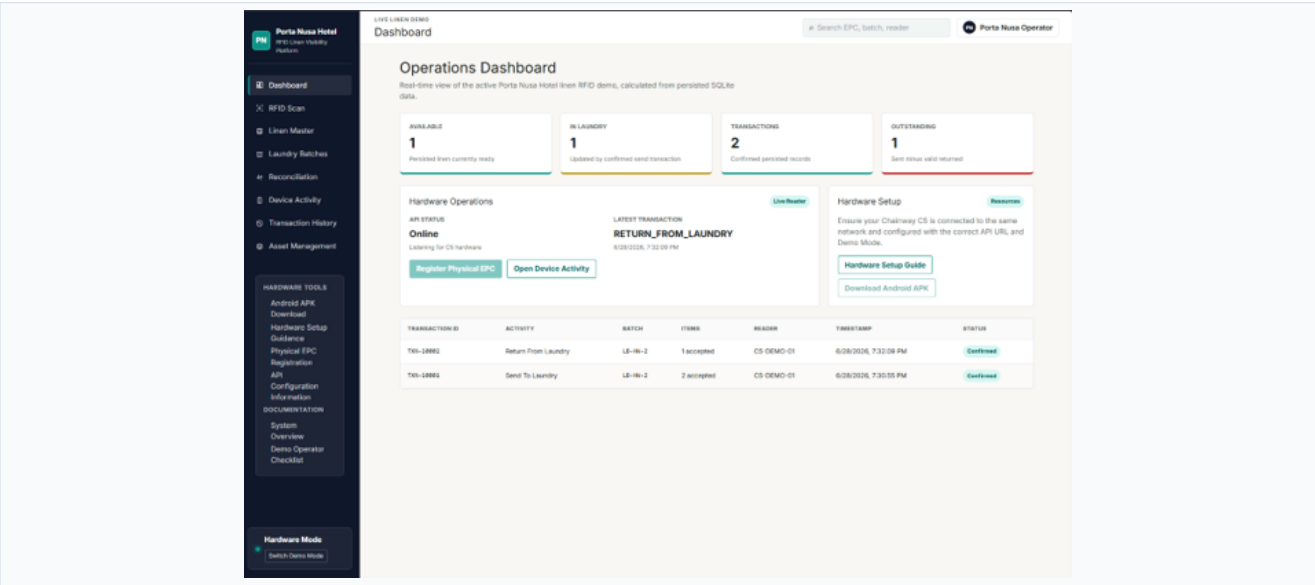
1	Register Physical EPC	Trigger a STOCK_COUNT scan with an unregistered tag in range. The API stores the EPC as UNKNOWN_EPC. The Linen Master page polls for unknown EPCs every ~2.5 seconds and displays them in a registration queue. The operator assigns a Linen Code and Type.
2	STOCK_COUNT	Scan registered linen items in a location. Recognised EPCs return ACCEPTED; unregistered EPCs return UNKNOWN_EPC. RSSI and read count are captured per tag.
3	SEND_TO_LAUNDRY	Operator enters a batch code on the C5 and scans dispatch items. The API finds or creates the batch using the exact submitted code. The Laundry Batches page shows the batch as In Progress.
4	Partial RETURN_FROM_LAUNDRY	Operator returns a subset of items. Outstanding = Sent – Returned. The Reconciliation page lists the batch with specific outstanding EPCs highlighted.
5	Reconciliation Review	Reconciliation shows all batches with outstanding > 0. The operator can identify which EPCs have not yet been returned before proceeding.
6	Final Return	Remaining outstanding items are scanned and returned. Outstanding drops to 0.
7	Completed Batch	When outstanding = 0 and acceptedSent > 0, batch status becomes Completed. Completed batches are retained in Laundry Batches but disappear from Reconciliation.
8	Empty Reconciliation	With no outstanding items across any batch, the Reconciliation page shows an empty state and the Dashboard Outstanding metric shows 0.

Idempotency: All write operations are idempotent by session ID. If the C5 retries an upload due to a network interruption, the server returns a success response without creating duplicate records. Transaction History shows each session only once.

5. Web Platform — Key Screens

Hardware Mode Dashboard

The Dashboard aggregates live counts from the active mode database. The screenshot below shows the hardware database state during a partial return: one item is available, one remains in laundry, and one is outstanding.

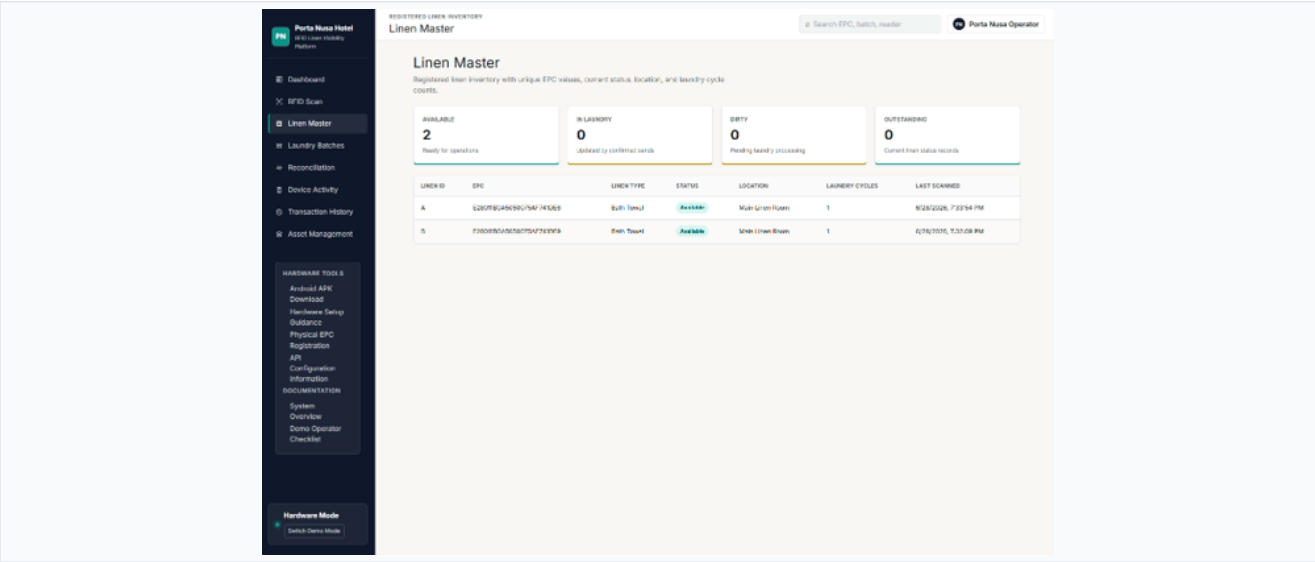


Hardware Mode Dashboard — partial return state. Available: 1, In Laundry: 1, Outstanding: 1, Transactions: 2. Latest transaction: RETURN_FROM_LAUNDRY.

Available	Items currently in stock and not in laundry
In Laundry	Items currently dispatched to laundry (outstanding > 0)
Outstanding	Total items sent to laundry but not yet returned
Transactions	Total RFID sessions recorded in the active database

Linen Master

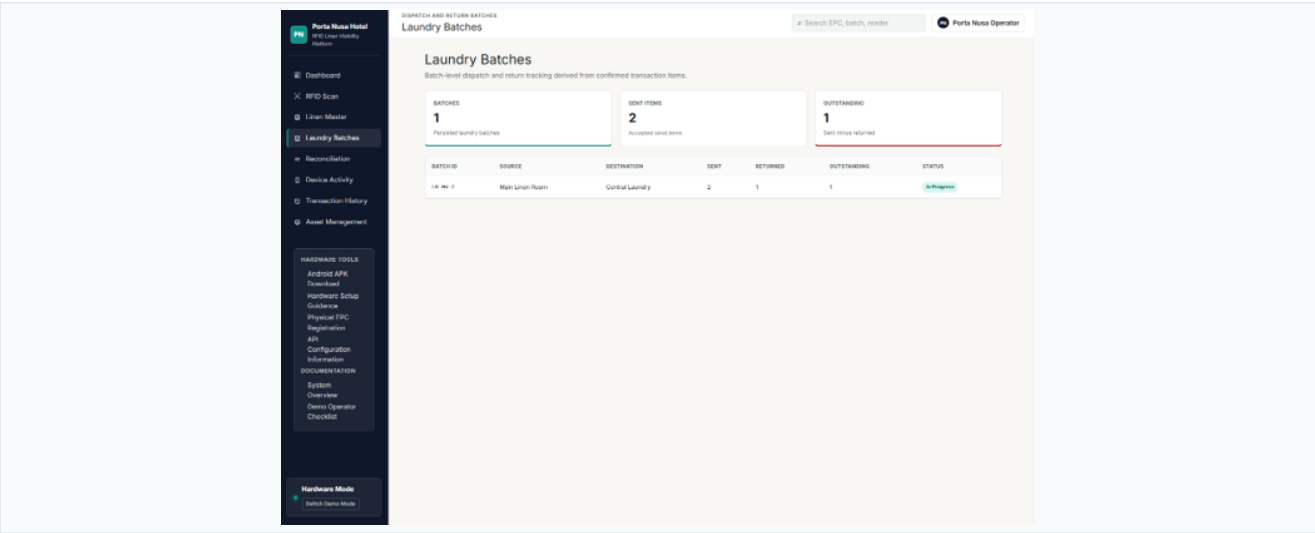
The Linen Master lists all registered linen items. In Hardware Mode, it also shows an auto-updating registration queue for unknown EPCs discovered by the C5 (polling every ~2.5 s).



Hardware Mode — Linen Master. Two registered linen items (IDs A and B), both showing Available status.

Laundry Batches

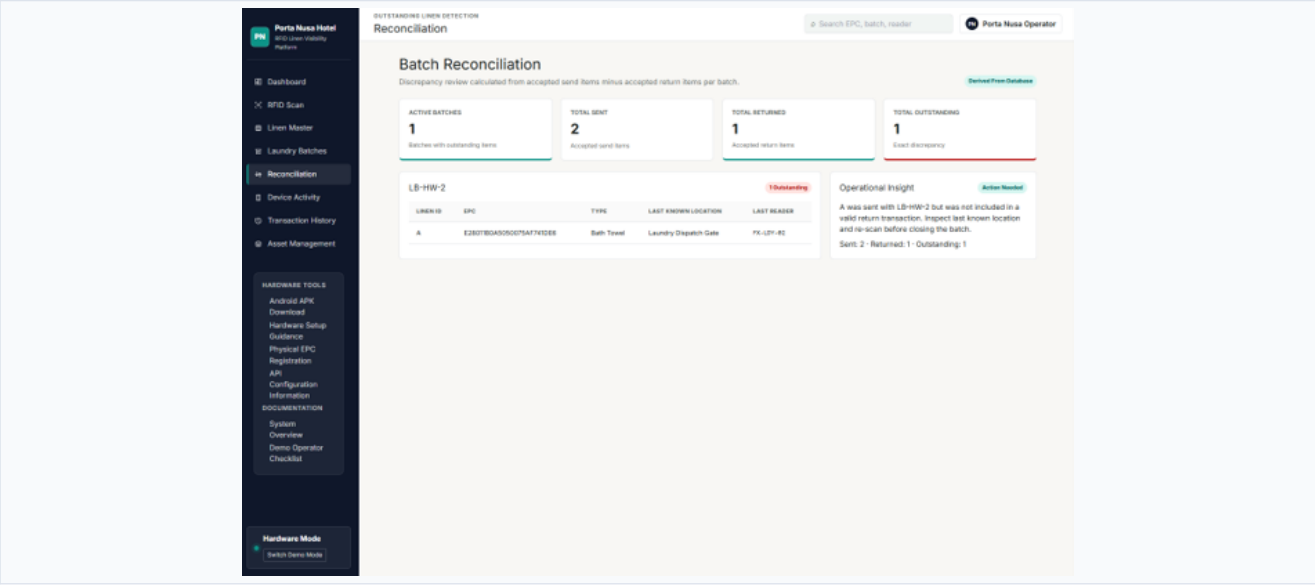
Laundry Batches lists all batches created through SEND_TO_LAUNDRY sessions. Outstanding is computed live from accepted sent minus accepted returned — never stored.



Hardware Mode — Laundry Batches. Batch LB-HW-2: Sent 2, Returned 1, Outstanding 1, Status: In Progress.

Reconciliation

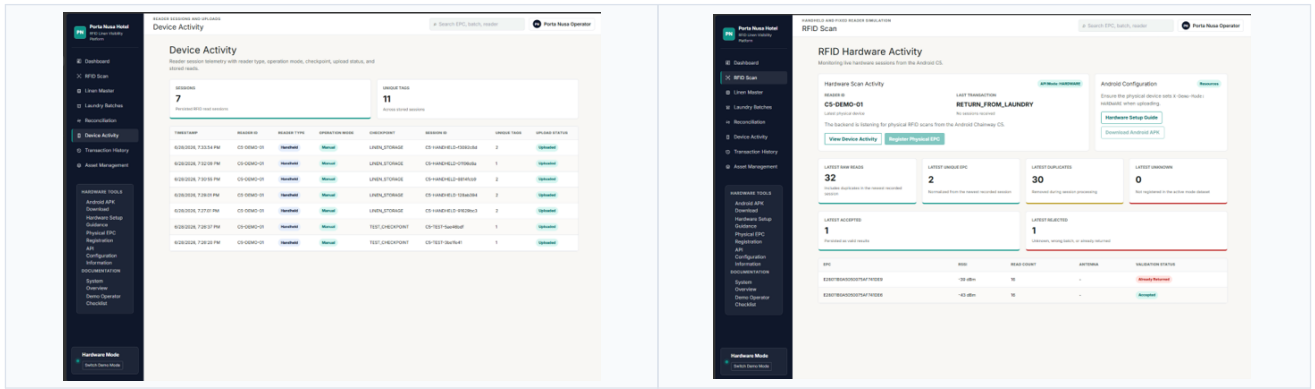
Reconciliation shows only batches with outstanding > 0. When a batch is fully returned its card disappears automatically. The screenshot below shows the In Progress state after a partial return — one item remains outstanding.



Hardware Mode — Reconciliation after a partial return. Batch LB-HW-2 remains In Progress: 2 items sent, 1 returned, 1 outstanding.

Device Activity and RFID Hardware Activity

Device Activity logs every reader session received from the C5. The RFID Scan page in Hardware Mode shows the latest recorded session data including per-EPC validation status — it is a read-only view, not a browser-side scanner.



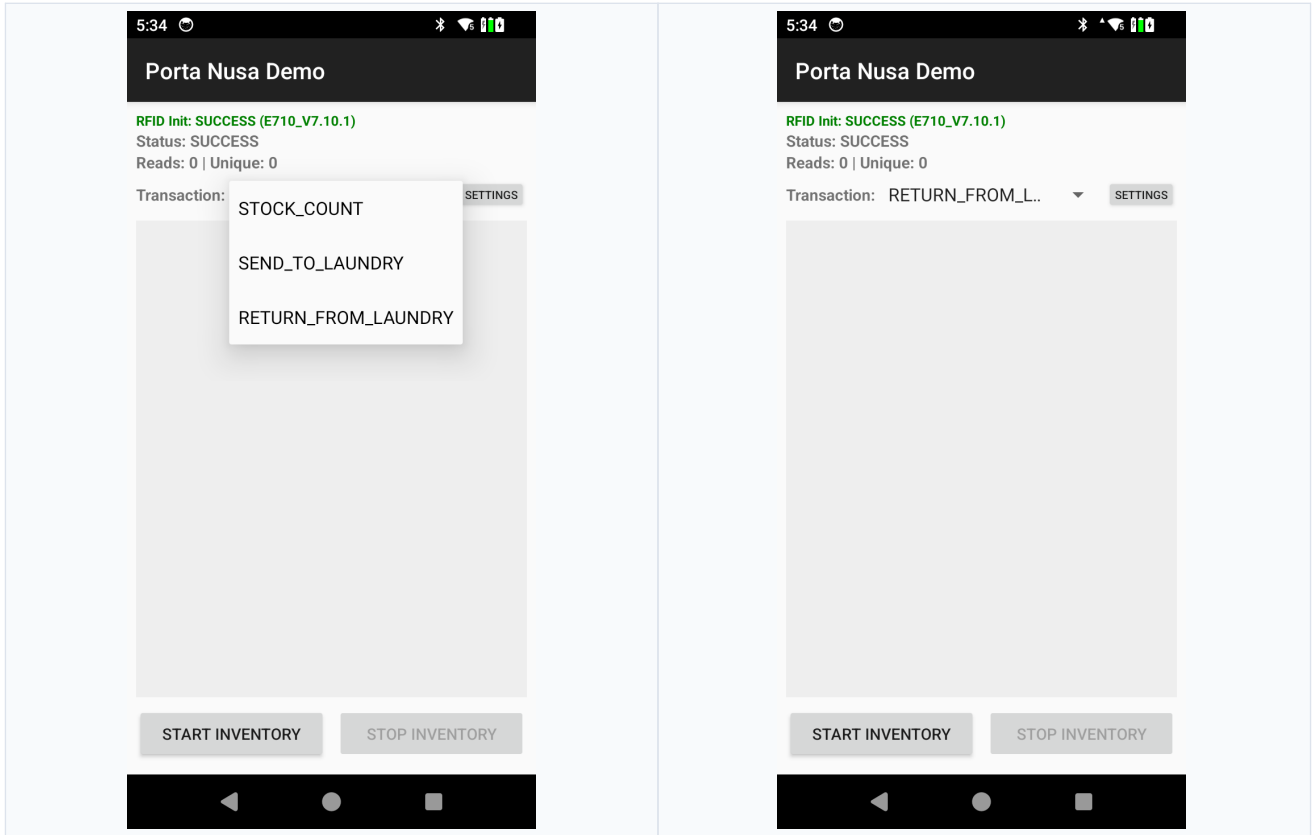
Device Activity — 7 reader sessions, 11 unique tags.

RFID Hardware Activity — last transaction: RETURN_FROM_LAUNDRY.

6. Chainway C5 Operation

The Chainway C5 E710 runs the Porta Nusa Operator Android app, which handles workflow selection, physical RFID scanning, and session upload to the web API.

Workflow Selection and Scanning



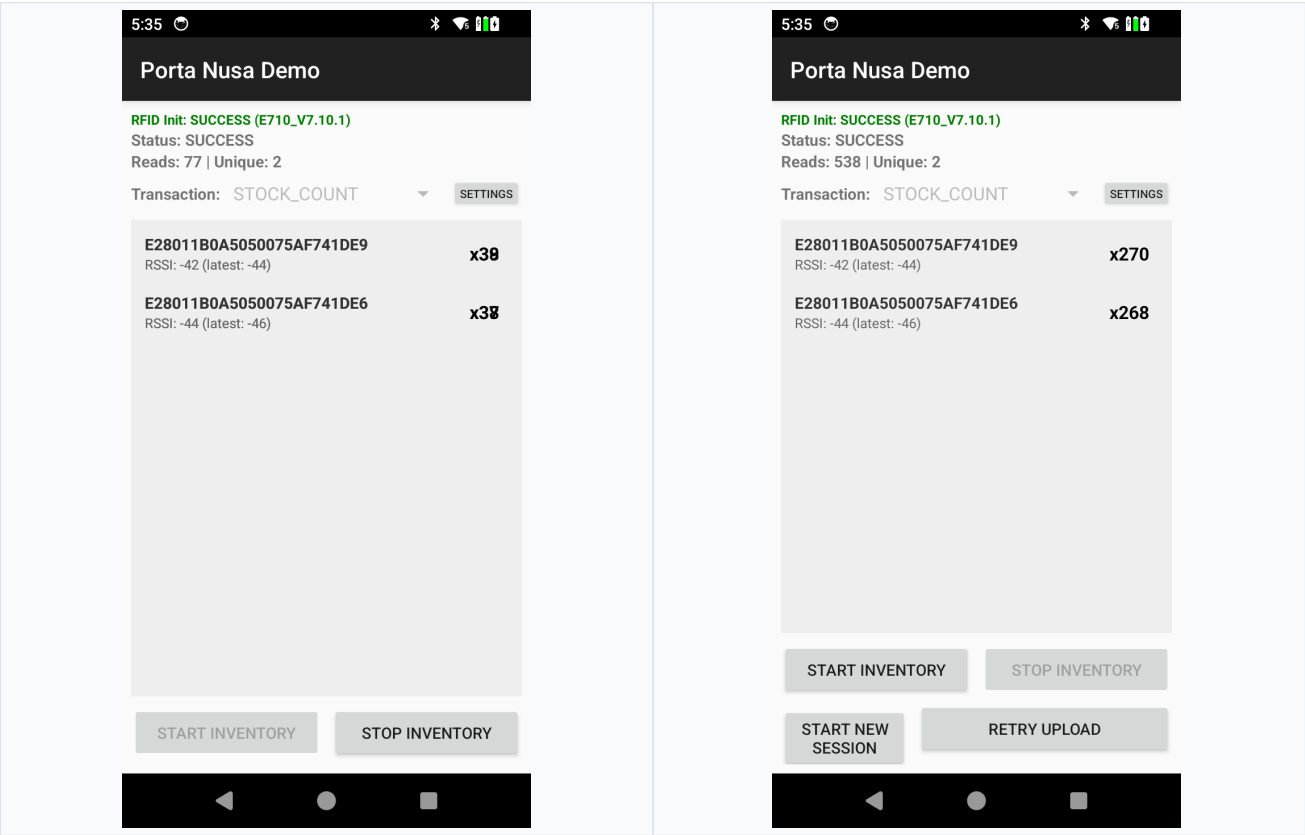
Workflow dropdown — STOCK_COUNT, SEND_TO_LAUNDRY, RETURN_FROM_LAUNDRY.

New session ready. Operator enters batch code before scanning.

The operator triggers a scan using the physical side key or the on-screen START SCAN button. The antenna reads all UHF RFID tags within the active power profile range. RSSI and read count are captured per tag. Duplicate EPC reads

within a single session are deduplicated — each unique EPC is submitted once per session.

Live Scan and Retry

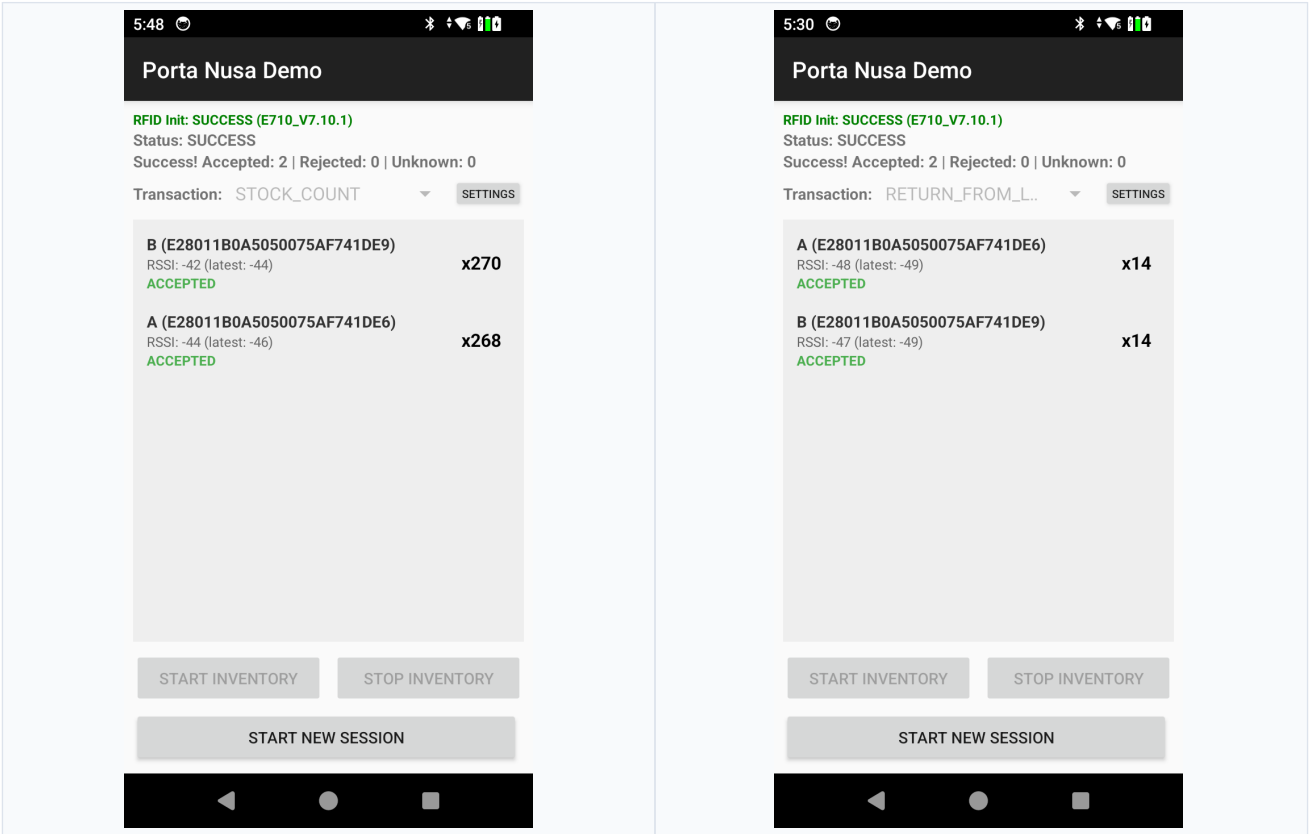


Live STOCK_COUNT scan. Reads: 77, Unique EPCs: 2.

Scan stopped — RETRY UPLOAD shown if initial upload failed.

Session Results

ACCEPTED	EPC recognised; batch matched; not previously processed.	No action needed.
WRONG_BATCH	EPC belongs to a different batch than submitted.	Confirm correct batch code and retry.
ALREADY_RETURNED	EPC was already returned in a prior session.	No action — item already reconciled.
UNKNOWN_EPC	EPC not registered in the hardware database.	Register via Linen Master before including in a workflow.

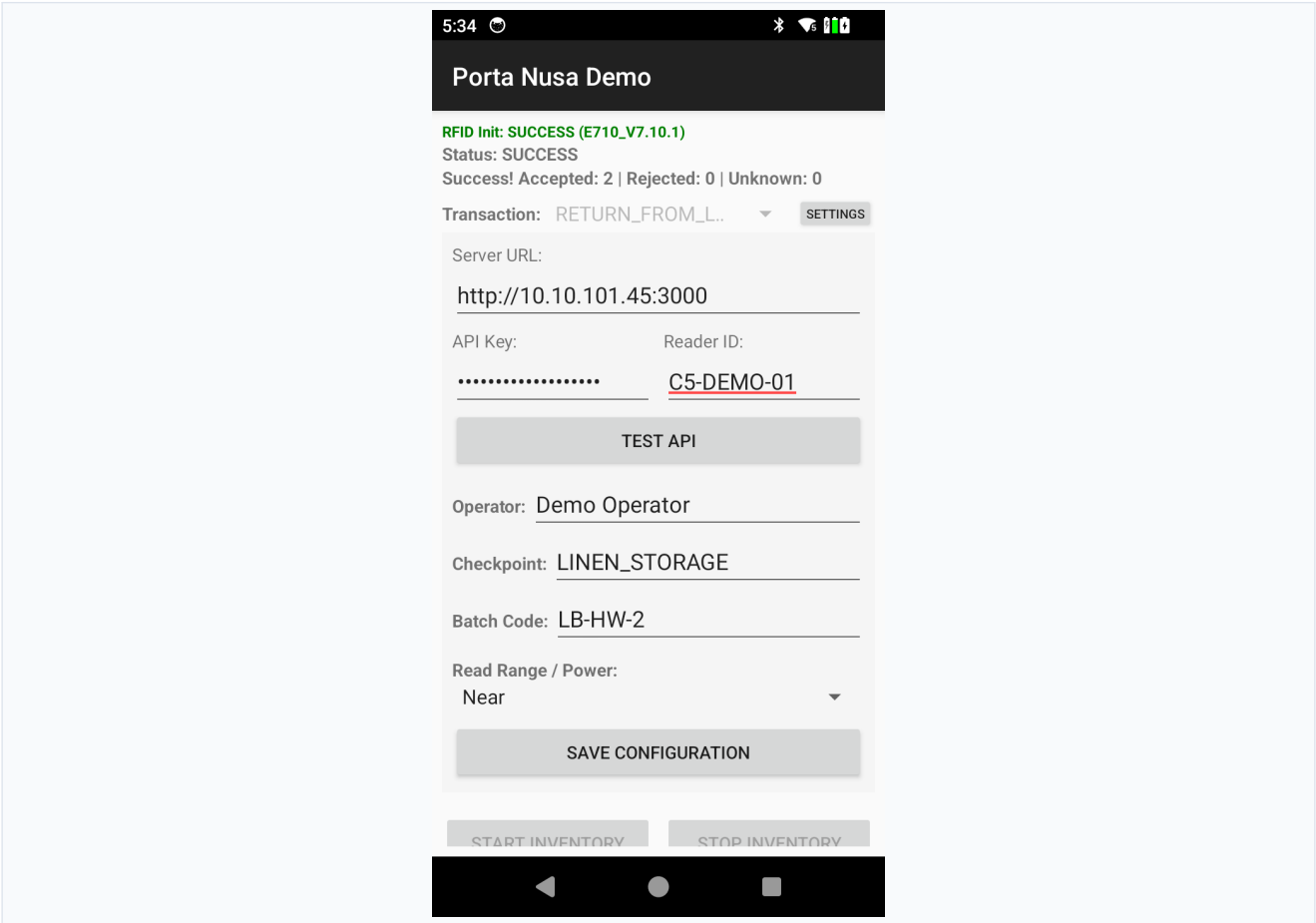


STOCK_COUNT result — Accepted: 2; both EPCs show ACCEPTED.

RETURN_FROM_LAUNDRY accepted result. Two EPCs were accepted; each EPC was read 14 times during the session.

C5 Settings

The Settings panel allows the operator to configure the server address, reader ID, batch code, and power profile. Settings persist across app restarts via SharedPreferences when the operator taps SAVE CONFIGURATION.



Chainway C5 Settings. The Server URL shown (<http://10.10.101.45:3000>) is a local demo environment address and must be reconfigured per deployment. Power Profile: Near shown.

7. RFID Power Profiles

The operator selects a power profile before each scan. The profile controls the power level applied to the UHF antenna, affecting how far the antenna can reliably read tags. Higher power gives wider coverage; lower power gives more precise, short-range reads.

Near	5	Single-item precision scanning. Minimises accidental reads from adjacent items.
Medium (default)	18	Standard room-level scanning. Balanced range and accuracy. Recommended for most operations.
Far	30	Wide-area coverage. Suitable for warehouse environments or bulk linen carts.

Implementation: The power value is passed directly to `setPower()` before every `startInventoryTag()` call. If `setPower()` fails or returns false, the scan is aborted and the operator sees 'Status: FAILED TO SET POWER'. This prevents silent reads at an unintended power level.

Persistence: The profile is saved only when the operator taps SAVE CONFIGURATION in the Android Settings panel. Changing the spinner without saving does not persist. Effective read distances depend on tag orientation, antenna position, and environmental conditions; the power values are configured levels, not guaranteed distances.

8. Simulation Mode User Guide

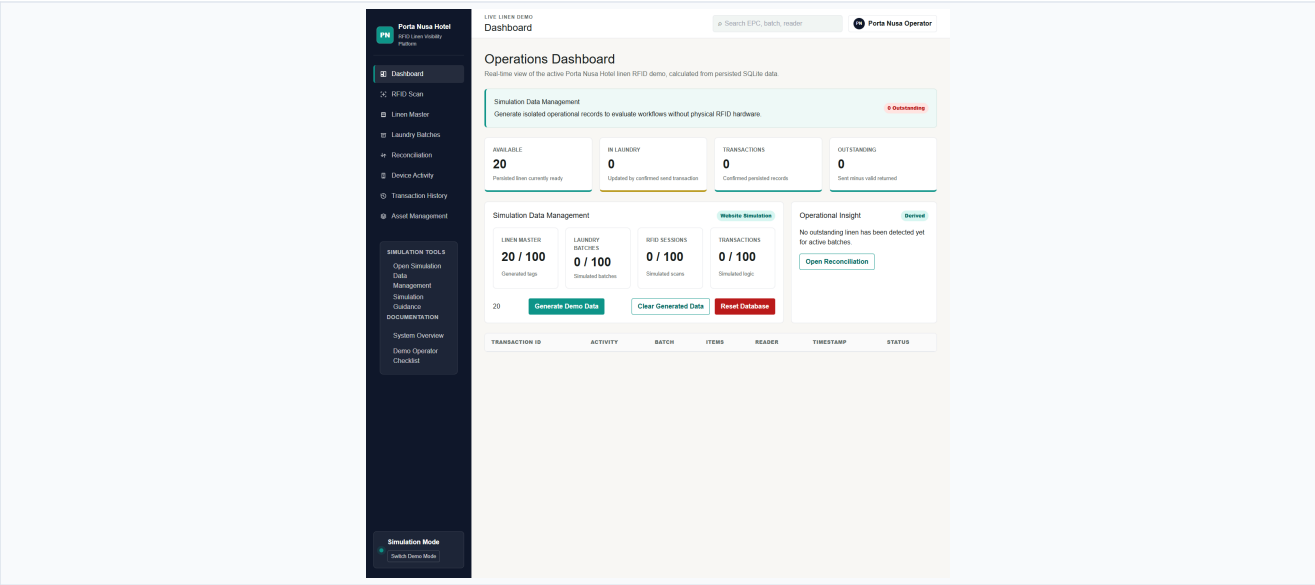
Complete all steps using the browser only — no physical hardware required.

Entering Simulation Mode

- 1. Open the platform in the browser.
- 2. If the sidebar shows **Hardware Mode**, click **Switch Demo Mode**.
- 3. Sidebar now shows **Simulation Mode**. All pages read from and write to the simulation database.

Simulation Data Management

The Dashboard contains the canonical **Simulation Data Management** section for all synthetic data operations.



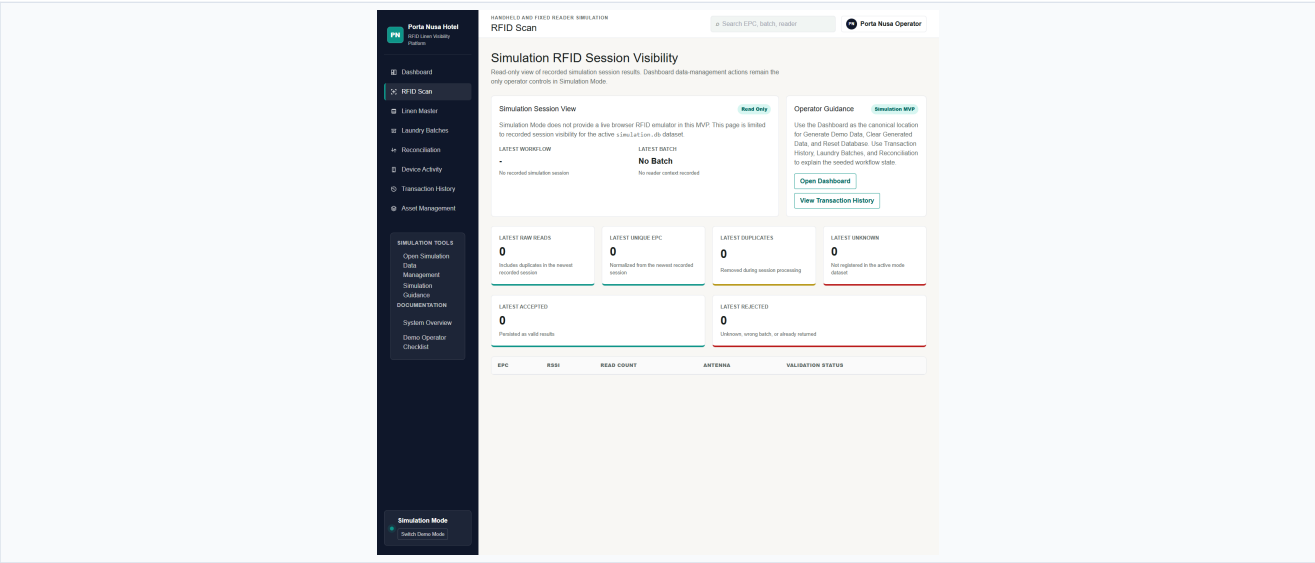
Simulation Mode Dashboard — Generate Demo Data, Clear Generated Data, and Reset Database actions visible.

Generate Demo Data	Creates synthetic linen items, batches, and transactions. Safe to run multiple times — capped at 100 items.
Clear Generated Data	Removes all synthetic records, returning the simulation database to an empty state.
Reset Database	Full wipe — restores only the base location records. Use for a guaranteed clean slate.

Pre-demo recommendation: Start on the Dashboard. Run Clear Generated Data, then Generate Demo Data. Use Reset Database only for a full wipe to base locations.

RFID Scan Page — Read-Only in Simulation Mode

The RFID Scan page in Simulation Mode is a read-only visibility page. It summarises the most recent recorded session data. It does not provide a live browser-side RFID scanner or transaction submission control.



Simulation Mode — RFID Scan Page. Read-only view of recorded session data. No live browser scan controls are provided.

Operator note: Do not present this page as a live browser RFID emulator. Browser operators manage seeded data from the Dashboard and review recorded outcomes here.

Recommended Simulation Demo Sequence

- 1. **Confirm mode:** Sidebar shows Simulation Mode.
- 2. **Reset data:** Dashboard → Clear Generated Data → Generate Demo Data.
- 3. **Dashboard:** Walk through metrics and the Simulation Data Management section.
- 4. **Linen Master:** Show seeded linen inventory.
- 5. **Laundry Batches and Reconciliation:** Explain the seeded batch state.
- 6. **Transaction History:** Show the recorded audit trail.
- 7. **RFID Scan:** Present as read-only evidence — not a live scanner.
- 8. **Close on Dashboard:** Restate Simulation vs Hardware database isolation.

9. Hardware Mode User Guide

Hardware Mode connects the Chainway C5 E710 to the web platform over a local network. Physical RFID scan sessions are uploaded by the Android app and processed by the Web API.

Pre-Requisites

Application service running	Browser opens the platform dashboard	Start the application service

C5 on same network	Wi-Fi icon visible on C5 status bar	Connect C5 to the same network as the server
Server address correct in app	C5 Settings shows correct address	Update Server URL in C5 Settings → SAVE CONFIGURATION
Browser in Hardware Mode	Sidebar shows Hardware Mode	Click Switch Demo Mode in the sidebar

Stale APK warning: Always ensure the installed APK was built from the current source. A stale APK may upload to an incorrect server or use an outdated session format.

Workflow Steps

1. **Confirm the application service is running** and the browser dashboard loads.
2. **Verify C5 network connectivity:** Confirm Wi-Fi is connected and the server address in C5 Settings is correct.
3. **Configure reader:** Select desired Power Profile (recommend Medium). Tap SAVE CONFIGURATION.
4. **Scan unknown EPC (new tags only):** Trigger STOCK_COUNT. Unknown EPC appears in the Linen Master queue within ~2.5 seconds.
5. **Register linen:** In Linen Master, assign a Linen ID and type. Click Register. The tag is now recognised.
6. **Run STOCK_COUNT:** Scan registered items. Confirm ACCEPTED results in Device Activity.
7. **Run SEND_TO_LAUNDRY:** Enter a batch code, scan dispatch items. Confirm batch is In Progress in Laundry Batches.
8. **Perform partial return:** Return a subset of items. Confirm Outstanding decreases in Reconciliation.
9. **Inspect Reconciliation:** Outstanding items visible by EPC. Batch remains In Progress.
10. **Perform final return:** Return remaining items. Outstanding drops to 0.
11. **Confirm completed state:** Laundry Batches shows Completed. Reconciliation shows empty state.

10. Demo Operator Checklist

Complete all items before and during a live demonstration.

Pre-Demo Checks

Service and Network

- Application service is running and the browser dashboard loads.
- Network is available and stable on the demo machine.
- Browser is confirmed in the correct starting mode (Simulation or Hardware).

Simulation Mode Data

- Sidebar shows Simulation Mode.
- Dashboard — Simulation Data Management section is visible.
- Clear Generated Data then Generate Demo Data — Dashboard shows non-zero item count.
- Laundry Batches and Reconciliation reflect the seeded state.

Hardware Mode (if using C5)

- Chainway C5 is powered on and the Porta Nusa Operator app is open.
- C5 Settings: server address points to the correct server.
- C5 Settings: Power Profile is set to the intended profile (recommend: Medium).
- SAVE CONFIGURATION tapped on C5.
- Test STOCK_COUNT scan performed — session appears in Device Activity within a few seconds.
- At least one linen item is registered in the hardware database.
- At least one RFID tag is available for physical demonstration.

Simulation Mode Demo Sequence

- Confirm sidebar shows Simulation Mode.
- Dashboard: run Clear Generated Data then Generate Demo Data if a fresh state is needed.
- Linen Master: confirm generated items are visible.
- Laundry Batches: explain the seeded batch state.
- Reconciliation: confirm it matches the seeded outstanding state.
- Transaction History: explain the recorded audit trail.
- RFID Scan: present as a read-only page — not a live scanner.
- Return to Dashboard: restate Simulation vs Hardware isolation.

Hardware Mode Demo Sequence

- Switch browser to Hardware Mode.
- Confirm C5 Settings show the correct server address and Power Profile.
- EPC Registration (new tags): STOCK_COUNT → unknown EPC in Linen Master queue → Register.
- STOCK_COUNT: scan registered items → ACCEPTED results in Device Activity.
- SEND_TO_LAUNDRY: enter batch code → scan dispatch items → batch In Progress.
- Power Profile demo (optional): switch Near / Far / Medium, show range difference.
- RETURN_FROM_LAUNDRY (partial): scan subset → Outstanding decreases in Reconciliation.
- RETURN_FROM_LAUNDRY (final): return remaining items → Outstanding = 0.
- Confirm Completed: Laundry Batches shows Completed. Reconciliation shows empty state.
- Transaction History: all sessions visible with timestamps and per-EPC results.

Fallback Procedures

C5 Unavailable: Switch to Simulation Mode and continue with the seeded dataset. Dashboard actions, inventory views, reconciliation, and transaction history remain demonstrable without physical hardware.

Application Service Unavailable: Confirm the service is still active. Restart if needed. If unavailable, demonstrate using pre-captured screenshots or a screen recording.

RFID Tags Not Being Read: Check Power Profile — switch to Medium or Far. Ensure tags are oriented horizontally. Restart the C5 app. If unresolved: switch to Simulation Mode.

Stop Conditions

STOP — Application crash or unrecoverable error	Pause, restart the service, and resume from the last completed checkpoint.
STOP — Results do not match expected behaviour	Stop immediately. Note the discrepancy. Do not claim the result is correct. Investigate after the session.
STOP — Hardware data visible in Simulation Mode (or vice versa)	Database isolation is a core guarantee. This is a critical integrity failure requiring investigation.

11. Troubleshooting

C5 shows upload error / connection refused	Confirm the application service is running. Verify the C5 server address is correct and both devices are on the same network.
Hardware scan data not appearing in browser	Confirm browser is in Hardware Mode. If data is still missing, the APK may be stale — reinstall from current source.
Unknown EPC not appearing in Linen Master queue	Confirm Hardware Mode is active. Wait up to 2.5 seconds. Hard-refresh the page if queue remains empty.
FAILED TO SET POWER on C5	Restart the app. Try a lower power profile (Near). If persistent, power-cycle the C5 device.
RETURN_FROM_LAUNDRY shows WRONG_BATCH	The batch code entered for return does not match the send batch code. Re-enter the exact code used during SEND_TO_LAUNDRY.
ALREADY_RETURNED on all returned items	Items were already returned in a prior session. Check Device Activity to confirm. No further action needed.
Reconciliation shows a batch that should be completed	Outstanding count is not yet zero — partial return recorded. Return remaining items; batch disappears when outstanding = 0.
SQLite lock during hardware database reset	Stop the application service and all processes accessing the hardware database before running the reset script. Restart the service after completion.
All results show UNKNOWN_EPC after EPC registration	Stale APK likely installed. Reinstall from current source on C5.
Unexpected page behaviour after source changes	The application build cache may be stale. Stop the service, clear the build cache, and restart.
Browser and C5 show different state for the same workflow	Browser may be in Simulation Mode while C5 targets Hardware Mode. Switch browser to Hardware Mode.
Scan reads far more items than expected	Power profile is set to Far. Switch to Near or Medium in C5 Settings and save.

12. Verification Matrix

The table below records the highest verification level achieved for each capability. DEPLOYMENT VERIFIED is not claimed — the platform has not been deployed to a production server.

Core Web UI (all routes)	BROWSER VERIFIED	All routes loaded in Chrome
Simulation Mode database isolation	BROWSER VERIFIED	Simulation database confirmed isolated
Hardware Mode database isolation	BROWSER VERIFIED	Hardware database confirmed isolated
STOCK_COUNT (physical)	PHYSICALLY VERIFIED	C5 upload confirmed with real RFID tags
SEND_TO_LAUNDRY (physical)	PHYSICALLY VERIFIED	Dynamic batch creation confirmed
RETURN_FROM_LAUNDRY (physical)	PHYSICALLY VERIFIED	WRONG_BATCH and ALREADY_RETURNED rejections confirmed
Dynamic batch reconciliation	PHYSICALLY VERIFIED	17/17 automated assertions + physical validation
Web-first EPC registration	PHYSICALLY VERIFIED	Unknown EPC appeared in browser queue within 2.5 s
RFID power profiles (Near/Medium/Far)	PHYSICALLY VERIFIED	Physical range comparison + persistence after restart
Retry / idempotency	AUTOMATED TESTED	17/17 assertions; no physical duplicate session test
In-app guide routes	BROWSER VERIFIED	All four guide routes loaded in Chrome
Simulation Data Management	BROWSER VERIFIED	Generate / Clear / Reset actions click-tested
Hardware database lifecycle	AUTOMATED TESTED	init, backup, reset, verify PASS on Windows
Production deployment	NOT VERIFIED	Deployment to production server not yet performed

Verification boundary: BROWSER VERIFIED confirms pages loaded and key interactions were tested in Chrome on a local development server. PHYSICALLY VERIFIED confirms the complete operation — physical RFID reads, C5 uploads, and correct data in the browser — was observed end-to-end. DEPLOYMENT VERIFIED is not claimed.

13. Known Limitations

The following are factual boundaries of the current MVP. Each is known and intentional for this phase of the project.

No production deployment

The platform runs on a local development server. Cloud deployment has been intentionally deferred. DEPLOYMENT VERIFIED is not claimed for any milestone.

Screenshots contain environment-specific demo data

Screenshots were captured from a local development environment. Item counts, EPC values, batch codes, and timestamps reflect that specific session and are not real hotel data.

Local endpoint values are environment-specific

The C5 Settings server address reflects the local development machine at time of capture. This value must be reconfigured per deployment environment.

No APK distribution portal

The Android APK must be sideloaded from a build workstation. There is no in-app distribution mechanism in this MVP.

No hardware database reset via the browser

Clearing hardware data requires the CLI reset script or manual database file replacement. Simulation data can be reset directly from the browser Dashboard.

No authentication layer

The platform has no login or multi-user session management. A single operator context is assumed per browser session.

Desktop browser required

The web UI is designed for desktop or tablet (minimum 1280 px width). Narrow mobile viewports may require horizontal scrolling.

Linen item cap at 100

Both databases enforce a hard cap of 100 registered linen items as a safe MVP limit.

Asset Management is an expansion concept

Asset Management is not a current core capability. The platform focuses on linen tracking through laundry cycles. Broader asset management is a planned expansion.

Some Android workflow screenshots not captured

SEND_TO_LAUNDRY accepted-result and WRONG_BATCH rejection screenshots were not captured to avoid risky hardware state mutation. These are optional cosmetic assets.

14. Glossary of Key Terms

The following terms appear throughout this document and in the web platform interface.

Batch Code	A short identifier (e.g. LB-HW-2) assigned to a laundry dispatch event. The operator enters this on the Chainway C5 before a SEND_TO_LAUNDRY scan. The API creates the batch automatically if it does not already exist.
EPC (Electronic Product Code)	The unique identifier stored in each UHF RFID tag. The C5 reads and reports the EPC of every tag in range. EPCs are registered against linen items in the Linen Master.
Hardware Mode	The operating mode in which all read/write operations target the hardware database. Used for live physical RFID operations with the Chainway C5 E710.
Hardware Database	The isolated SQLite database that stores hardware-mode linen items, batches, transactions, and device sessions. No data is shared with the simulation database.
Outstanding	The computed count of items sent to laundry that have not yet been returned. Formula: acceptedSent – validReturned, computed live — never stored.
Power Profile	A named configuration (Near, Medium, Far) for the UHF antenna power level. The value is passed directly to the RFID SDK setPower() call before each scan.
Reconciliation	The web platform page that lists all active batches with outstanding > 0. A batch card disappears automatically when its outstanding count reaches 0.

RFID (Radio-Frequency Identification)	Technology used to identify linen items. UHF RFID tags are read by the Chainway C5 E710 handheld reader over distances of centimetres to several metres depending on the power profile and tag orientation.
Simulation Mode	The operating mode in which all read/write operations target the simulation database. No physical hardware is required. Demo data is managed from the Dashboard.
Simulation Database	The isolated SQLite database for simulation-mode data. Synthetic linen items and transactions are generated and cleared from the Dashboard's Simulation Data Management section.
STOCK_COUNT	A workflow type that records a snapshot of which EPC tags are present at a location. Does not create or modify laundry batches.
SEND_TO_LAUNDRY	A workflow type that dispatches linen items to a laundry batch. Creates the batch automatically if the submitted batch code does not already exist.
RETURN_FROM_LAUNDRY	A workflow type that records the return of linen items from laundry. Outstanding decreases by each ACCEPTED return. ALREADY_RETURNED prevents double-counting.
X-Demo-Mode Header	HTTP request header injected by browser middleware (from the demoMode cookie) or hardcoded by the Chainway C5 app. Determines which database the API targets. Values: SIMULATION or HARDWARE.